

TESTS FOR CURED PROPORTION FOR RECURRENT EVENT COUNT DATA WHEN THE DATA ARE CENSORED

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Summary: Sumathi and Rao (2008) proposed a cure model for recurrent event count data. The proposed model was based on the zero inflated Poisson (ZIP) distribution. Several tests were proposed for testing the cured proportion when the data set is uncensored (Sumathi and Rao, 2010). But in long term follow up studies, the investigators very often come across situations where the patients are lost to follow up and the data sets are said to be censored. The present paper is an extension of the work of Sumathi and Rao (2010). In the present paper, tests are proposed for testing the cured proportion in a randomly censored recurrent event count data. The small sample performances of the proposed tests are studied using simulations.

1. Introduction

The widely used model in survival analysis is the Cox (1972) proportional hazards (PH) model. This model is based on the assumption that every individual in the population under study is susceptible to an adverse event such as a disease, and will eventually experience this event if the follow up is sufficiently long. The recent advancement in medical research has shown that a large proportion of the diseased population is being cured of various chronic diseases such as epilepsy and cancer. The commonly used Cox (1972) PH model cannot be used here because of its assumption. This led to the development of cure models.

The cure models are survival models basically developed to estimate the proportion of patients cured from cancer of mouth and throat, cervix, uteri and breast. Using these models, the investigator can also study the causes for the failure of the treatment in the uncured group of patients

and obtain the trends in the survival of patients suffering from cancer. The two cure models that are extensively used are the mixture model developed by Boag (1949) and the bounded cumulative hazard (BCH) model proposed by Yakovlev, Asselain, Bardou, Fourquet, Hoang, Rochefediere and Tsodikov (1993).

The mixture model and the BCH model are applicable only when the study variable is the time to occurrence of an adverse event. The models do not consider the possibilities of the number of recurrences of the event. For example, diseases like epilepsy and malaria could recur several times during the life span of an individual. Sumathi and Rao (2008) proposed a cure model for recurrent event count data. The proposed model turned out to be zero inflated Poisson (ZIP) model. Several tests were proposed for testing the cured proportion in a recurrent event count data when the data are uncensored. But in long term follow up studies, the investigators very often come across data sets which are censored, i.e., data sets where the patients are lost to follow up. This paper provides an extension of the work of Sumathi and Rao (2010) by proposing tests for the cured proportion where there is random censoring.

For a dataset with a continuous failure time, random censoring can be dealt without much difficulty. However, there is no standard way of analyzing a recurrent event count data set in which the observations are censored. Clayton (1994) has suggested that the censored observations can be treated as missing observations and can be analyzed using the techniques available for missing data.

In the present paper, a new method is proposed for handling censored observations in a recurrent event count data set. With an assumption that the censoring time t_i for the i^{th} individual is known, the number of recurrences is said to follow independent inflated Poisson process with parameters p and λt_i , $i = 1, 2, \dots, n$, where n denotes the number of individuals who are under follow up. This enables us to propose conditional likelihood for the parameters p and λ , given the censoring times. Using the conditional likelihood, conditional tests are proposed for testing the cured proportion p .

The rest of the paper is organized as follows. With a brief discussion on the estimation of the parameters, the asymptotic normality of the conditional maximum likelihood estimators (MLEs) of the parameters are proved in section 2. Section 3 discusses the marginal inference for the cured proportion. The marginal likelihood is explicitly derived when the censoring time follows Weibull distribution. In section 4, tests are proposed for testing the cured proportion. These tests are an extension of the tests proposed in Sumathi and Rao (2010) to the case of random censoring. Section 5 discusses the small sample performances of the tests based on the estimated type I error rates. The power comparisons of the tests are discussed in section 6. The small sample comparisons indicate that a variant of the Wald test emerges as the best in terms of maintaining type I error rate and the power function. The paper concludes with a discussion in section 7.

2. Conditional Inference for Cured Proportion

Sumathi and Rao (2008) proposed a ZIP model as a cure model for recurrent event count data. The following notations are used in the sequel with reference to the ZIP model.

A random variable Y is said to follow a zero inflated Poisson distribution with parameters p and

λ if its probability mass function (p. m. f.) is given by

$$P[Y = y] = \begin{cases} p + (1 - p) \exp(-\lambda), & \text{when } y = 0 \\ (1 - p) \exp(-\lambda) \frac{\lambda^y}{y!}, & \text{when } y = 1, 2, \dots \end{cases} \quad (1)$$

Here, p and λ denote the inflate parameter (cured proportion) and the mean parameter, respectively. Let n denote the number of individuals who are under a follow up. The follow up period is sufficiently long. Let T be the follow up period. Further, let $Y_1, Y_2, \dots, Y_n$ denote the number of recurrences of the disease of the n individuals under study and let $\tilde{t}_1, \tilde{t}_2, \dots, \tilde{t}_n$ be their censoring times, with cumulative distribution function $F(\tilde{t}_i; \cdot)$. The study is restricted to the case of random censoring. Let $t_i = \min(\tilde{t}_i, T)$, $i = 1, 2, \dots, n$. When t_i 's, $i = 1, 2, \dots, n$, are known, as in the uncensored case, we can assume that Y_i 's follow inflated Poisson process with parameters p and λt_i , $i = 1, 2, \dots, n$, where λ denotes the mean recurrence rate per unit time. Then the conditional likelihood for p and λ can be written as

$$L(p, \lambda \mid \underline{t}; \underline{Y}) = \prod_{i=1}^n \left(p + (1 - p) e^{-\lambda t_i} \right)^{\delta_i} \left(\frac{(1 - p) e^{-\lambda t_i} (\lambda t_i)^{y_i}}{y_i!} \right)^{(1 - \delta_i)} \quad (2)$$

where $\underline{t} = (t_1, t_2, \dots, t_n)$, $\underline{Y} = (y_1, y_2, \dots, y_n)$ and $\delta_i = \begin{cases} 1, & y_i = 0 \\ 0, & y_i \neq 0 \end{cases}$

The minimal sufficient statistics for p and λ are given by δ and $\sum_{i=1}^n y_i$, respectively. Here, $\delta = (\delta_1, \dots, \delta_n)$.

The maximum likelihood (ML) equations for p and λ are

$$\sum_{i=1}^n \frac{\delta_i (1 - e^{-\lambda t_i})}{p + (1 - p) e^{-\lambda t_i}} - \frac{n - n_0}{1 - p} = 0. \quad (3)$$

$$\text{and, } \sum_{i=1}^n \frac{-\delta_i t_i (1 - p) e^{-\lambda t_i}}{p + (1 - p) e^{-\lambda t_i}} + \sum_{i=1}^n (1 - \delta_i) \left(\frac{y_i}{\lambda} - t_i \right) = 0, \text{ respectively.} \quad (4)$$

The conditional score function for λ given δ is

$$\sum_{i=1}^n (1 - \delta_i) \left(\frac{y_i}{\lambda} - \frac{t_i}{(1 - e^{-\lambda t_i})} \right) \quad (5)$$

This equation is independent of p and is the optimal function for estimating λ ignoring p (Kale, 1998). This result is similar to the case of uncensored setup where no optimal function exists for estimating p , ignoring λ .

For an uncensored data set, the ML equation for λ is

$$\lambda - \left(\frac{\sum_{y_i \neq 0} y_i}{n - n_0} \right) (1 - e^{-\lambda}) = 0.$$

(See Sumathi and Rao (2010)). This equation is independent of p . Moreover, this ML equation is the same as the conditional score function equated to zero and thus the MLE $\hat{\lambda}$ is the optimal estimator of λ . In the present case of censored data setup, the ML equation (4) given above, for λ , cannot be simplified. This equation is not independent of p . Thus, unlike the uncensored case (Sumathi and Rao, 2010), the MLE of λ in the model for the censored data, is not the optimal estimator.

For carrying out inference on p , the determination of the restricted MLE of λ (denoted by $\hat{\lambda}$) when $p = p_0$ (specified) is necessary. $\hat{\lambda}$ is the solution of the restricted ML equation $\sum_{i=1}^n \frac{-\delta_i t_i (1-p_0) e^{-\lambda t_i}}{p_0 + (1-p_0) e^{-\lambda t_i}} + \sum_{i=1}^n (1-\delta_i) \left(\frac{y_i}{\lambda} - t_i \right) = 0$.

The conditional expected Fisher information matrix is $F(p, \lambda | \tilde{T}) = \begin{bmatrix} F_{11} & F_{12} \\ F_{12} & F_{22} \end{bmatrix}$ (6)

where $F_{11} = \sum_{i=1}^n \frac{(1-e^{-\lambda t_i})^2}{p+(1-p)e^{-\lambda t_i}} + \frac{1}{1-p} \sum_{i=1}^n (1-e^{-\lambda t_i})$,

$F_{12} = -\sum_{i=1}^n t_i e^{-\lambda t_i} - \sum_{i=1}^n \frac{t_i e^{-\lambda t_i} (1-e^{-\lambda t_i})(1-p)}{p+(1-p)e^{-\lambda t_i}}$ and

$F_{22} = (1-p)^2 \sum_{i=1}^n \frac{(t_i e^{-\lambda t_i})^2}{p+(1-p)e^{-\lambda t_i}} + \frac{1-p}{\lambda} \sum_{i=1}^n (1-e^{-\lambda t_i}) \frac{y_i}{\lambda} - (1-p) \sum_{i=1}^n (t_i^2 e^{-\lambda t_i})$.

The following theorem establishes the asymptotic normality of $(\hat{p}, \hat{\lambda})'$, where $\hat{p}$ and $\hat{\lambda}$ are the MLEs of p and λ , respectively and are the solutions to the ML equations given by (3) and (4) above.

Theorem: Under the regularity conditions of Czado and Min (2005) suitably modified, the following holds.

$$F^{\frac{1}{2}}(p, \lambda | \tilde{T})' \left(\begin{pmatrix} \hat{p} \\ \hat{\lambda} \end{pmatrix} - \begin{pmatrix} p \\ \lambda \end{pmatrix} \right) \xrightarrow{D} N_2(O, I_2),$$

where $F^{\frac{1}{2}}(p, \lambda | T)$ is the Cholesky square root matrix of the Fisher information matrix given by (6) and $N_2(O, I_2)$ denotes the bivariate normal distribution with mean vector O and variance-covariance matrix I_2 . I_2 is an identity matrix of order 2. $\tilde{T}$ is the random variable denoting the censoring time.

Proof: Follows from the theorem of Czado and Min (2005). The mean parameter μ in their theorem is replaced by $\lambda t_i = \exp(\alpha + \log(t_i))$ where $\log(t_i)$ acts as a single covariate with a known value of the regression coefficient equal to 1.

3. Marginal Inference for Cured Proportion

Given the conditional likelihood (2), the marginal likelihood function for p and λ is given by

$$L(p, \lambda, \theta; Y) = \int L(p, \lambda | \tilde{T}; Y) f(t_i; \theta) dt_i \quad (7)$$

where $f(t_i; \theta)$ is the density of the censoring variable $\tilde{T}$ and $\int$ corresponds to a multiple integral. In general, no closed form solution exists for the marginal likelihood function given by (7). In the appendix, the expressions for the full marginal likelihood function for p and λ , when the censoring time $\tilde{T}$ follows Weibull distribution are derived.

When the censoring time $\tilde{T}$ follows exponential distribution, for an infinite follow up period i.e., $T = \infty$, the full marginal likelihood is

$$L(p, \lambda, \theta; y) = \prod_{i=1}^n \frac{(\theta + p\lambda)^{\delta_i} (\theta(1-p)\lambda^{y_i})^{1-\delta_i}}{(\lambda + \theta)^{y_i(1-\delta_i)+1}}. \quad (8)$$

A close look at the expression (8) reveals that this is the likelihood for the inflated geometric distribution with parameter $\frac{\theta}{\lambda + \theta}$. Further, it can be observed that the dimension for the minimal

sufficient statistics $(n_0, \sum_{i=1}^n y_i)$ is lower than the number of parameters (p, λ, θ) and thus, one can estimate p and $\frac{\theta}{\lambda + \theta}$, but not λ and θ separately. When the follow up period T is finite, the marginal likelihood function does not reduce to a closed form for either the Weibull or the exponential distributions. In the present paper, the attention has been restricted to the conditional inference only.

4. Tests for cured proportion

Just as has been done in the uncensored data scheme (Sumathi and Rao, 2010), the cured proportion p in a censored data setup is estimated and a confidence interval for the same can be developed. Since the confidence interval can be obtained by the acceptance region of a test (Lehmann and Romano, 2005), tests are proposed for testing the cured proportion. To be specific, we shall test for the null hypothesis $H_0 : p = p_0$ against the two sided alternative hypothesis $H_1 : p \neq p_0$. Thirteen tests have been proposed and the test statistics are as follows.

4.1. Likelihood Ratio Test (LRT)

The LRT statistic for testing the null hypothesis $H_0 : p = p_0$ is

$W_1 = 2 \left(\log L(\hat{p}, \hat{\lambda}; \underline{y}) - \log L(p_0, \hat{\lambda}; \underline{y}) \right)$, which simplifies to

$$W_1 = 2 \sum_{i=1}^n \left[\delta_i \log \frac{\hat{p} + (1 - \hat{p}) e^{-\hat{\lambda} t_i}}{p_0 + (1 - p_0) e^{-\hat{\lambda} t_i}} + (1 - \delta_i) \left\{ \log \frac{(1 - \hat{p})}{(1 - p_0)} - (\hat{\lambda} - \hat{\lambda}) t_i + y_i \log \frac{\hat{\lambda}}{\hat{\lambda}} \right\} \right]. \quad (9)$$

The asymptotic null distribution of W_1 is central chi-squared with one degree of freedom. The signed LRT statistic for testing one sided alternative is

$$W_2 = \text{sgn} \left((\hat{p} - p_0) \sqrt{W_1} \right). \quad (10)$$

4.2. Wald Test

The Wald test statistic for testing the null hypothesis $H_0 : p = p_0$ is

$$W_3 = \frac{(\hat{p} - p_0)}{SE(\hat{p})} \quad (11)$$

where $SE(\hat{p})$ denotes the standard error of $\hat{p}$ which can be obtained from the inverse of the Fisher information matrix F . The asymptotic null distribution of W_3 is standard normal.

4.3. Score Test

Rao and Bartlett (1948) proposed the score test. The advantage of this test is that when the hypothesis is simple, the computation of the MLEs is not required. Details of the score test are available in Rao (1973), Cox and Hinkley (1974) and Severini (2000). The score test statistic is

$$W_4 = U^1 F^{-1} U \quad (12)$$

where

$$U = \left[\frac{\partial}{\partial p} \log L; \frac{\partial}{\partial \lambda} \log L \right] \quad (13)$$

is the score vector and F is the Fisher information matrix given by (6). The parameters p and λ in U and F are replaced by p_0 and $\hat{\lambda}$, respectively. The score functions in U are

$$\frac{\partial \log L}{\partial p} = \sum_{i=1}^n \left(\delta_i \frac{1 - e^{-\lambda t_i}}{p + (1-p)e^{-\lambda t_i}} + (1 - \delta_i) \left(\frac{-1}{(1-p)} \right) \right) \quad (14)$$

$$\frac{\partial \log L}{\partial \lambda} = \sum_{i=1}^n \left(\frac{-\delta_i t_i (1-p)e^{-\lambda t_i}}{p + (1-p)e^{-\lambda t_i}} + (1 - \delta_i) \left(\frac{y_i}{\lambda} - t_i \right) \right) \quad (15)$$

where L is given by (2). The asymptotic null distribution of W_4 is central chi-squared with one degree of freedom. For the one sided alternative hypothesis, the signed score test statistic, which has the same form as the signed LRT given by (10), can be proposed. Asymptotically, the LRT, Wald and the score tests are equivalent under the null as well as the alternative hypotheses. The results follow from the standard asymptotics for the LRT, Wald and the score tests. For details, see Cox and Hinkley (1974) and Severini (2000). The small sample performances of these tests differ. The results of the simulations discussed in section 5 indicate the same.

4.4. Variants of the Wald and Score Tests

The standard Wald test uses the estimates $\hat{p}$ and $\hat{\lambda}$ in the Fisher information matrix F to obtain the $SE(\hat{p})$ while the standard score test uses p_0 and $\hat{\lambda}$. The values of p_0 and $\hat{\lambda}$, and p_0 and $\hat{\lambda}$ may also be used in the Fisher information matrix F for the computation of the Wald test statistic. For details, see Cox and Hinkley (1974) and Kale (1999). Similarly, p_0 and $\hat{\lambda}$, and $\hat{p}$ and $\hat{\lambda}$ may be used in F for the computation of the score test statistic. The three versions of the score test which include the standard score test are examined by Morgan, Palmer and Ridout (2007). Let the variants of the Wald test statistic be denoted as W_5 and W_6 that of the score test statistic be W_7 and W_8 , respectively.

The test statistics W_5 and W_6 have the same forms as that of the conventional Wald test statistic W_3 given by (11) except that in the expression for $SE(\hat{p})$ of W_3 , we substitute p_0 and $\hat{\lambda}$ for p and λ , respectively in F to obtain W_5 , and p_0 and $\hat{\lambda}$ in F to obtain W_6 . Similarly, in the score test statistic W_4 , we replace p and λ in F by p_0 and $\hat{\lambda}$, respectively to obtain W_7 and, by $\hat{p}$ and $\hat{\lambda}$ to obtain W_8 . Again, these test statistics W_7 and W_8 have the same forms as that of the conventional score test statistic W_4 given by (12).

4.5. New Variants of the Score test

Motivated by the work of Sumathi and Rao (2010), six new variants of the score test are proposed in addition to the tests proposed above and are given below.

$$W_9 = \left(\frac{\partial}{\partial p_0} \log L \left(p_0; \hat{\lambda}, y \right) \right) F_{p,p}^{-1/2} \left(p_0; \hat{\lambda} \right) \quad (16)$$

$$W_{10} = \left(\frac{\partial}{\partial p_0} \log L \left(p_0; \hat{\lambda}, y \right) \right) F_{p,p}^{-1/2} \left(p_0; \hat{\lambda} \right) \quad (17)$$

$$W_{11} = \left(\frac{\partial}{\partial p_0} \log L \left(p_0; \hat{\lambda}, y \right) \right) F_{p,p}^{-1/2} \left(p_0; \hat{\lambda} \right) \quad (18)$$

$$W_{12} = \left(\frac{\partial}{\partial p_0} \log L \left(p_0; \hat{\lambda}, y \right) \right) F_{p,p}^{-1/2} \left(p_0; \hat{\lambda} \right) \quad (19)$$

$$W_{13} = \left(\frac{\partial}{\partial p_0} \log L \left(p_0; \hat{\lambda}, y \right) \right) F_{p,p}^{-1/2} \left(\hat{p}_0; \hat{\lambda} \right) \quad (20)$$

$$W_{14} = \left(\frac{\partial}{\partial p_0} \log L \left(p_0; \hat{\lambda}, y \right) \right) F_{p,p}^{-1/2} \left(\hat{p}_0; \hat{\lambda} \right) \quad (21)$$

4.6. Small sample type I error rates

As has been done in the case of uncensored data scheme (Sumathi and Rao, 2010), the performances of only thirteen test statistics, viz., $W_1, W_3, W_4, W_5, W_6, W_7, W_8, W_9, W_{10}, W_{11}, W_{12}, W_{13}$ and W_{14} were assessed based on a simulation study. The test W_2 has not been dealt with since it is the test statistic for one sided alternative, while in this paper, only the two sided alternative $H_1 : p \neq p_0$ is considered. The maximum follow up period T was fixed to be 60 months. The percentage of censoring C was considered to be 20. The actual follow up time $\tilde{t}_i$ of the i^{th} individual was considered to be a uniform random variable. The parameter of the uniform distribution was determined so as to yield 20% censoring.

The remaining simulation configurations were as follows. The level of significance α was 5%. Samples of sizes 20, 40, 80, 100, 200 and 400 were drawn. The hypothesis was tested when the values of the cured proportion p_0 were 0.1, 0.3, 0.5, 0.7 and 0.9. Testing was also done when p_0 assumed values 0 and 1. The mean parameter λ with values 1, 3, 5, 7 and 9 were considered. The number of simulations carried out was 10000. The value of λ was chosen so as to represent the number of teeth decayed, missing or filled given by the DMFT index. The details regarding this data can be found in Sumathi and Rao (2008).

The observations were generated from a zero inflated Poisson process with parameters p_0 and λt_i where $t_i = \min(\tilde{t}_i, T)$, $i = 1, 2, \dots, n$. The values of the test statistics were computed and the number of times the hypothesis was rejected was noted to estimate the type I error rates. The observed Fisher information matrix was used for simulations since the practitioners are more likely to use the observed Fisher information matrix rather than the expected Fisher information matrix while testing. For the zero inflated Poisson distribution considered in the present paper, the observed conditional Fisher information matrix is given by

$$O(p, \lambda | \tilde{T}) = \begin{bmatrix} o_{11} & o_{12} \\ o_{12} & o_{22} \end{bmatrix} \quad (22)$$

$$\text{where } o_{11} = \sum_{i=1}^n \frac{\delta_i (1 - e^{-\lambda t_i})^2}{(p + (1-p)e^{-\lambda t_i})^2} + \frac{1}{(1-p)^2} \sum_{i=1}^n (1 - \delta_i),$$

$$o_{12} = - \sum_{i=1}^n \frac{\delta_i t_i e^{-\lambda t_i}}{p + (1-p)e^{-\lambda t_i}} - \sum_{i=1}^n \frac{\delta_i t_i e^{-\lambda t_i} (1 - e^{-\lambda t_i}) (1-p)}{(p + (1-p)e^{-\lambda t_i})^2} \text{ and}$$

$$o_{22} = (1-p)^2 \sum_{i=1}^n \frac{\delta_i (t_i e^{-\lambda t_i})^2}{(p + (1-p)e^{-\lambda t_i})^2} + \frac{1}{\lambda^2} \sum_{i=1}^n (1 - \delta_i) y_i - (1-p) \sum_{i=1}^n \frac{\delta_i t_i^2 e^{-\lambda t_i}}{(p + (1-p)e^{-\lambda t_i})}.$$

The lower and the upper $\frac{\alpha^{\text{th}}}{2}$ percentiles of the distributions of the test statistics were also recorded.

The performances of the tests with reference to the attained level of significance ($\hat{\alpha}$) are evaluated as follows. A test is said to maintain type I error rate if $\hat{\alpha}$ is in the interval $[0.04, 0.06]$ i.e., $\hat{\alpha} \pm 0.01$. The test is liberal if $\hat{\alpha} > 0.06$ and conservative, if $\hat{\alpha} < 0.04$. The test becomes too conservative if $\hat{\alpha} < 0.02$ (i.e., 60% below the nominal level α). When two tests under consideration fail to maintain type I error rates, the test which is conservative is preferred to a test which is liberal.

In the following paragraphs, comparisons between the tests are made in the order of their performances with reference to their ability to maintain type I error rates. Tables 1, 2 and 3 give the estimated values of type I error rates of the tests when the sample sizes are 20, 40 and 80, respectively for all combinations of p and λ that have been considered. Table 4 summarizes the results of the simulations performed to estimate the type I error rates of the various tests. This table gives the number of times each of the test statistics maintain type I error rate, the number of times the test becomes conservative and the number of times they become liberal, corresponding to various sample sizes for a total of 25 configurations of p and λ considered. The percentages are given within brackets.

For a sample of size 20, the LRT W_1 maintains type I error rate at 11 (44%) configurations out of 25. It maintains the same for all values of λ when testing for $p_0 = 0.5$ and 0.7. The test is conservative for the remaining 14 (56%) configurations. The variants of the Wald test viz., W_5 and W_6 maintain type I error rates at 10 (40%) configurations but are liberal at 15 (60%) configurations. These tests maintain type I error rate for all values of λ when testing for $p_0 = 0.3$ and 0.5. The score test W_4 and its variants $W_7, W_8, W_9, W_{10}, W_{11}, W_{12}, W_{13}$, and W_{14} maintain type I error rate at 5 (20%) configurations whereas the Wald test W_3 fails to maintain even once. The Wald test becomes conservative at 5 (20%) configurations and is liberal at the remaining 20 (80%).

When the sample size is increased to 40, the LRT W_1 continues to stand first with an increase in the number of times it maintains type I error rate. The test maintains type I error rate at 15 (60%) configurations for all values of λ considered when testing for $p_0 = 0.3, 0.5$ and 0.7, becomes conservative at 5 (20%) configurations and is liberal at the remaining 5 (20%). The variants of the Wald test viz., W_5 and W_6 also maintain type I error rate at 15 (60%) configurations but are liberal at the remaining 10 (40%) and hence have been rated second. The variants of the score test viz., W_8, W_{13} and W_{14} maintain type I error rates at 12 (48%) of the configurations. The Wald's test W_3 maintains type I error rates at 11 (44%) configurations out of 25. It maintains type I error rates

for all values of λ testing for $p_0 = 0.5$ and 0.7 . The test is conservative for the remaining 14 (56%) configurations. The score test W_4 , its variant W_7 and the four new variants W_9 , W_{10} , W_{11} and W_{12} maintain type I error rates at 10 (40%) configurations. All these tests maintain type I error rate for all values of λ when testing for $p_0 = 0.5$ and 0.7 . The tests are conservative at 10 (40%) configurations i.e., for all values of λ considered when testing for $p_0 = 0.1$ and 0.3 , and liberal at 5 (20%) configurations for all values of λ considered when testing for $p_0 = 0.9$.

When the sample size is increased to 80, the variants of the Wald test viz., W_5 and W_6 and the variants of the score test viz., W_8 , W_{13} and W_{14} maintain type I error rate at 20 (80%) configurations. These tests are liberal at the remaining 5 (20%). The tests are liberal for all values of λ considered when testing for $p_0 = 0.9$. The LRT W_1 , the score test W_4 and its variants W_7 , W_9 , W_{10} , W_{11} and W_{12} maintain type I error rates at 18 (72%) configurations. The LRT W_1 is liberal at 2 (8%) configurations while the score test W_4 and its variants W_7 , W_9 , W_{10} , W_{11} and W_{12} are liberal at 7 (28%) configurations. The Wald test W_3 maintains type I error rate at 16 (64%) configurations and is liberal at the remaining 9 (36%) configurations. The proposed tests maintain type I error rate while testing for $p_0 = 0.3$, 0.5 and 0.7 for all values λ of considered. The performances of the tests when the sample sizes are 100 and 200 are similar to that when the sample size is 80 but for the sample size $n = 400$, all the tests maintain type I error rate for all combinations of p_0 and λ considered.

5. Small Sample Power Comparisons

Having assessed the behavior of the tests with respect to the type I error rates, the power of the tests have also been studied as has been done in the uncensored case by Sumathi and Rao (2010). The simulation configurations are similar to those presented in the previous section except that the observations are generated from the values of p and λ under the alternative hypothesis. The criteria for assessment of the tests based on the powers are as follows.

- (a) Power of the test in the neighborhood of the null hypothesis $H_0 : p = p_0$ for a fixed value of λ and
- (b) The rate at which the power function attains the value 1.

Although the power functions have been estimated for all the tests and configurations, the comparisons have been restricted to only those cases where the tests maintain type I error rates. Since most of the tests maintain type I error rate at $p_0 = 0.3$ and 0.5 , the power comparisons of the tests are reported only for values of $p_0 = 0.3$ and 0.5 when $\lambda = 3$ and 5 . The power functions of the tests for these configurations are given in figures 1 to 12 for the sample sizes $n = 20$, 40 and 80 .

For the sample size $n = 20$, the powers of some of the tests are low (< 0.1) for alternatives in the neighborhood of p_0 and the power curves of the tests are not distinguishable in this region. Moreover, the score test and some of its variants exhibit fluctuations in the power functions (decreasing or randomly decreasing and increasing after a certain stage) rather than increasing on either sides of the specified value of the inflate parameter. (For details, see Sumathi and Rao (2010, 2011)). From the graphs 1 to 12, it is evident that the Wald test W_3 and its variants viz., W_5 and W_6 have more power on either sides of $H_0 : p = p_0$. The score test W_4 and its variants W_7 , W_9 , W_{10} , W_{11} and W_{12} follow second in line. However, the power functions of the score test W_4 and its variants exhibit fluctuations

for sample sizes 20 and 40, as in the case of uncensored data set up. Thus a moderate sample of size 100 is required for the application of the score test and its variants. The LRT stands third with regard to its performance based on power. The variants of the score test W_8 , W_{13} and W_{14} behave alike with very low powers (< 0.1). Tables 5, 6 and 7 give the powers of the tests when testing for $p_0 = 0.3$ when $\lambda = 3$ for sample sizes 20, 40 and 80.

As mentioned earlier, the rate of convergence of the power functions to the value 1 also indicate the performance of the tests which are at a moderate or large distance from the null hypothesis. Here again it was observed that the Wald's test W_3 stands first followed by its variants W_5 and W_6 . The rate of convergence of the power function to the value 1 is again slow for the LRT W_1 , the score test W_4 and its variants when compared to the former three tests. This is also evident in figures 1 to 12 and tables 5, 6 and 7. Moreover, the tests have larger power on the left of p_0 than at an equidistant value of p on its right.

Table 1: Estimated Type I error rates of the tests when the sample size was $n = 20$.

λ	p_0	W_1	W_3	W_4	W_5	W_6	W_7	W_8	W_9	W_{10}	W_{11}	W_{12}	W_{13}	W_{14}
1	0.1	0.0165	0.0131	0.0098	0.1454	0.1454	0.0098	0.1435	0.0098	0.0098	0.0098	0.0098	0.1431	0.1431
	0.3	0.0203	0.0660	0.0648	0.0509	0.0509	0.0648	0.0477	0.0648	0.0648	0.0648	0.0648	0.0477	0.0477
	0.5	0.0418	0.0771	0.0418	0.0418	0.0418	0.0418	0.0115	0.0418	0.0418	0.0418	0.0418	0.0115	0.0115
	0.7	0.0539	0.0753	0.0658	0.0828	0.0828	0.0658	0.0829	0.0658	0.0658	0.0658	0.0658	0.0829	0.0829
	0.9	0.0470	0.0701	0.0814	0.1635	0.1653	0.0710	0.1960	0.0814	0.0711	0.0711	0.0711	0.1975	0.1975
3	0.1	0.0120	0.0076	0.0059	0.1483	0.1483	0.0059	0.1477	0.0059	0.0059	0.0059	0.0059	0.1477	0.1477
	0.3	0.0196	0.0674	0.0670	0.0493	0.0493	0.0670	0.0480	0.0670	0.0670	0.0670	0.0670	0.0480	0.0480
	0.5	0.0362	0.0806	0.0376	0.0362	0.0362	0.0376	0.0102	0.0376	0.0376	0.0376	0.0376	0.0102	0.0102
	0.7	0.0537	0.0707	0.0666	0.0835	0.0835	0.0666	0.0840	0.0666	0.0666	0.0666	0.0666	0.0840	0.0840
	0.9	0.0257	0.0763	0.0868	0.1460	0.1460	0.0822	0.2171	0.0868	0.0825	0.0825	0.0825	0.2304	0.2304
5	0.1	0.0135	0.0068	0.0062	0.1577	0.1577	0.0062	0.1573	0.0062	0.0062	0.0062	0.0062	0.1573	0.1573
	0.3	0.0179	0.0692	0.0690	0.0503	0.0503	0.0690	0.0496	0.0690	0.0690	0.0690	0.0690	0.0496	0.0496
	0.5	0.0397	0.0829	0.0408	0.0396	0.0396	0.0408	0.0103	0.0408	0.0408	0.0408	0.0408	0.0103	0.0103
	0.7	0.0516	0.0714	0.0697	0.0806	0.0806	0.0697	0.0811	0.0697	0.0697	0.0697	0.0697	0.0812	0.0812
	0.9	0.0184	0.0757	0.0886	0.1455	0.1456	0.0847	0.2135	0.0886	0.0858	0.0858	0.0858	0.2413	0.2413
7	0.1	0.0129	0.0085	0.0082	0.1478	0.1478	0.0082	0.1475	0.0082	0.0082	0.0082	0.0082	0.1475	0.1475
	0.3	0.0186	0.0731	0.0731	0.0519	0.0519	0.0731	0.0512	0.0731	0.0731	0.0731	0.0731	0.0512	0.0512
	0.5	0.0418	0.0770	0.0359	0.0418	0.0418	0.0359	0.0128	0.0359	0.0359	0.0359	0.0359	0.0128	0.0128
	0.7	0.0508	0.0685	0.0672	0.0799	0.0799	0.0672	0.0802	0.0672	0.0672	0.0672	0.0672	0.0802	0.0802
	0.9	0.0153	0.0751	0.0889	0.1360	0.1360	0.0843	0.2085	0.0889	0.0860	0.0860	0.0860	0.2411	0.2411
9	0.1	0.0120	0.0083	0.0082	0.1458	0.1458	0.0082	0.1454	0.0082	0.0082	0.0082	0.0082	0.1454	0.1454
	0.3	0.0200	0.0668	0.0668	0.0517	0.0517	0.0668	0.0513	0.0668	0.0668	0.0668	0.0668	0.0513	0.0513
	0.5	0.0413	0.0816	0.0393	0.0413	0.0413	0.0393	0.0110	0.0393	0.0393	0.0393	0.0393	0.0110	0.0110
	0.7	0.0518	0.0699	0.0693	0.0820	0.0820	0.0693	0.0826	0.0693	0.0693	0.0693	0.0693	0.0827	0.0827
	0.9	0.0151	0.0735	0.0871	0.1355	0.1355	0.0828	0.2119	0.0871	0.0854	0.0854	0.0854	0.2468	0.2468

Table 2: Estimated Type I error rates of the tests when the sample size was $n = 40$.

λ	p_0	W_1	W_3	W_4	W_5	W_6	W_7	W_8	W_9	W_{10}	W_{11}	W_{12}	W_{13}	W_{14}
1	0.1	0.0240	0.0051	0.0042	0.1014	0.1014	0.0042	0.0972	0.0042	0.0042	0.0042	0.0042	0.0972	0.0972
	0.3	0.0535	0.0234	0.0196	0.0421	0.0421	0.0196	0.0369	0.0196	0.0196	0.0196	0.0196	0.0369	0.0369
	0.5	0.0423	0.0439	0.0418	0.0418	0.0418	0.0418	0.0390	0.0418	0.0418	0.0418	0.0418	0.0390	0.0390
	0.7	0.0567	0.0522	0.0522	0.0413	0.0413	0.0520	0.0333	0.0522	0.0521	0.0521	0.0521	0.0333	0.0333
	0.9	0.0807	0.0350	0.1208	0.1640	0.1641	0.0665	0.1650	0.1208	0.0683	0.0683	0.0683	0.1651	0.1651
3	0.1	0.0174	0.0033	0.0031	0.1052	0.1052	0.0031	0.1048	0.0031	0.0031	0.0031	0.0031	0.1047	0.1047
	0.3	0.0525	0.0211	0.0194	0.0378	0.0378	0.0194	0.0317	0.0194	0.0194	0.0194	0.0194	0.0317	0.0317
	0.5	0.0376	0.0374	0.0361	0.0376	0.0376	0.0361	0.0366	0.0361	0.0361	0.0361	0.0361	0.0366	0.0366
	0.7	0.0603	0.0535	0.0540	0.0435	0.0435	0.0540	0.0378	0.0540	0.0540	0.0540	0.0540	0.0378	0.0378
	0.9	0.0868	0.0176	0.1245	0.1722	0.1722	0.0880	0.1781	0.1245	0.1013	0.1013	0.1013	0.1784	0.1784
5	0.1	0.0175	0.0038	0.0036	0.1038	0.1038	0.0036	0.1033	0.0036	0.0036	0.0036	0.0036	0.1033	0.1033
	0.3	0.0534	0.0185	0.0178	0.0406	0.0406	0.0178	0.0348	0.0178	0.0178	0.0178	0.0178	0.0348	0.0348
	0.5	0.0410	0.0400	0.0395	0.0409	0.0409	0.0395	0.0402	0.0395	0.0395	0.0395	0.0395	0.0402	0.0402
	0.7	0.0597	0.0516	0.0522	0.0443	0.0443	0.0521	0.0367	0.0522	0.0522	0.0522	0.0522	0.0367	0.0367
	0.9	0.0870	0.0093	0.1177	0.1687	0.1687	0.0772	0.1777	0.1177	0.1050	0.1050	0.1050	0.1788	0.1788
7	0.1	0.0190	0.0025	0.0024	0.1051	0.1051	0.0024	0.1049	0.0024	0.0024	0.0024	0.0024	0.1049	0.1049
	0.3	0.0560	0.0190	0.0185	0.0399	0.0399	0.0185	0.0345	0.0185	0.0185	0.0185	0.0185	0.0345	0.0345
	0.5	0.0359	0.0423	0.0416	0.0359	0.0359	0.0416	0.0358	0.0416	0.0416	0.0416	0.0416	0.0358	0.0358
	0.7	0.0565	0.0506	0.0508	0.0416	0.0416	0.0508	0.0346	0.0508	0.0508	0.0508	0.0508	0.0346	0.0346
	0.9	0.0851	0.0068	0.1224	0.1692	0.1692	0.0793	0.1784	0.1224	0.1119	0.1119	0.1119	0.1801	0.1801
9	0.1	0.0199	0.0021	0.0020	0.1120	0.1120	0.0020	0.1119	0.0020	0.0020	0.0020	0.0020	0.1119	0.1119
	0.3	0.0537	0.0202	0.0198	0.0391	0.0391	0.0198	0.0328	0.0198	0.0198	0.0198	0.0198	0.0328	0.0328
	0.5	0.0393	0.0414	0.0413	0.0393	0.0393	0.0413	0.0389	0.0413	0.0413	0.0413	0.0413	0.0389	0.0389
	0.7	0.0561	0.0517	0.0523	0.0422	0.0422	0.0522	0.0358	0.0523	0.0523	0.0523	0.0523	0.0358	0.0358
	0.9	0.0843	0.0055	0.1260	0.1680	0.1680	0.0819	0.1773	0.1260	0.1168	0.1168	0.1168	0.1799	0.1799

Table 3: Estimated Type I error rates of the tests when the sample size was $n = 80$.

λ	p_0	W1	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
1	0.1	0.0293	0.0542	0.0439	0.0596	0.0596	0.0439	0.0562	0.0439	0.0439	0.0439	0.0439	0.0562	0.0562
	0.3	0.0520	0.0545	0.0521	0.0488	0.0488	0.0521	0.0458	0.0521	0.0521	0.0521	0.0521	0.0458	0.0458
	0.5	0.0535	0.0557	0.0534	0.0533	0.0533	0.0534	0.0485	0.0534	0.0534	0.0534	0.0534	0.0485	0.0485
	0.7	0.0567	0.0561	0.0560	0.0447	0.0447	0.0560	0.0437	0.0560	0.0560	0.0560	0.0560	0.0437	0.0437
	0.9	0.0641	0.1030	0.0949	0.0942	0.0942	0.0945	0.0942	0.0949	0.0945	0.0945	0.0945	0.0942	0.0942
3	0.1	0.0311	0.0733	0.0641	0.0601	0.0601	0.0641	0.0594	0.0641	0.0641	0.0641	0.0641	0.0594	0.0594
	0.3	0.0473	0.0489	0.0481	0.0457	0.0457	0.0481	0.0447	0.0481	0.0481	0.0481	0.0481	0.0447	0.0447
	0.5	0.0586	0.0592	0.0584	0.0584	0.0584	0.0584	0.0565	0.0584	0.0584	0.0584	0.0584	0.0565	0.0565
	0.7	0.0539	0.0533	0.0533	0.0432	0.0432	0.0533	0.0432	0.0533	0.0533	0.0533	0.0533	0.0432	0.0432
	0.9	0.0618	0.0984	0.0904	0.0870	0.0870	0.0901	0.0871	0.0904	0.0902	0.0902	0.0902	0.0872	0.0872
5	0.1	0.0297	0.0665	0.0585	0.0572	0.0572	0.0585	0.0570	0.0585	0.0585	0.0585	0.0585	0.0570	0.0570
	0.3	0.0541	0.0549	0.0541	0.0479	0.0479	0.0541	0.0473	0.0541	0.0541	0.0541	0.0541	0.0473	0.0473
	0.5	0.0562	0.0568	0.0562	0.0562	0.0562	0.0562	0.0548	0.0562	0.0562	0.0562	0.0562	0.0548	0.0548
	0.7	0.0525	0.0524	0.0524	0.0443	0.0443	0.0524	0.0443	0.0524	0.0524	0.0524	0.0524	0.0443	0.0443
	0.9	0.0663	0.1021	0.0962	0.0952	0.0952	0.0961	0.0957	0.0962	0.0961	0.0961	0.0961	0.0957	0.0957
7	0.1	0.0269	0.0652	0.0574	0.0545	0.0545	0.0574	0.0542	0.0574	0.0574	0.0574	0.0574	0.0542	0.0542
	0.3	0.0513	0.0518	0.0513	0.0449	0.0449	0.0513	0.0442	0.0513	0.0513	0.0513	0.0513	0.0442	0.0442
	0.5	0.0542	0.0545	0.0540	0.0540	0.0540	0.0540	0.0531	0.0540	0.0540	0.0540	0.0540	0.0531	0.0531
	0.7	0.0472	0.0470	0.0470	0.0408	0.0408	0.0470	0.0408	0.0470	0.0470	0.0470	0.0470	0.0408	0.0408
	0.9	0.0620	0.0963	0.0899	0.0885	0.0885	0.0898	0.0886	0.0899	0.0898	0.0898	0.0898	0.0886	0.0886
9	0.1	0.0301	0.0720	0.0641	0.0613	0.0613	0.0641	0.0612	0.0641	0.0641	0.0641	0.0641	0.0612	0.0612
	0.3	0.0493	0.0496	0.0492	0.0430	0.0430	0.0492	0.0427	0.0492	0.0492	0.0492	0.0492	0.0427	0.0427
	0.5	0.0550	0.0554	0.0550	0.0550	0.0550	0.0550	0.0539	0.0550	0.0550	0.0550	0.0550	0.0539	0.0539
	0.7	0.0502	0.0500	0.0500	0.0435	0.0435	0.0500	0.0435	0.0500	0.0500	0.0500	0.0500	0.0435	0.0435
	0.9	0.0608	0.0980	0.0915	0.0851	0.0851	0.0915	0.0853	0.0915	0.0915	0.0915	0.0915	0.0853	0.0853

Table 4: Summary of the simulations (Percentages are given within brackets in each cell).

Sample size n	Feature of Type I error rate	W_1	W_3	W_4	W_5	W_6	W_7	W_8	W_9	W_{10}	W_{11}	W_{12}	W_{13}	W_{14}
20	Maintains	11(44)	00(00)	05(20)	10(40)	10(40)	05(20)	05(20)	05(20)	05(20)	05(20)	05(20)	05(20)	05(20)
	Conservative	14(56)	05(20)	05(20)	00(00)	00(00)	05(20)	05(20)	05(20)	05(20)	05(20)	05(20)	05(20)	05(20)
	Liberal	00(00)	20(80)	15(60)	15(60)	15(60)	15(60)	15(60)	15(60)	15(60)	15(60)	15(60)	15(60)	15(60)
40	Maintains	15(60)	11(44)	10(40)	15(60)	15(60)	10(40)	12(48)	10(40)	10(40)	10(40)	10(40)	12(48)	12(48)
	Conservative	05(20)	14(56)	10(40)	00(00)	00(00)	10(40)	03(12)	10(40)	10(40)	10(40)	10(40)	03(12)	03(12)
	Liberal	05(20)	00(00)	05(20)	10(40)	10(40)	05(20)	10(40)	05(20)	05(20)	05(20)	05(20)	10(40)	10(40)
80	Maintains	18(72)	16(64)	18(72)	20(80)	20(80)	18(72)	20(80)	18(72)	18(72)	18(72)	18(72)	20(80)	20(80)
	Conservative	05(20)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)
	Liberal	02(08)	09(36)	07(28)	05(20)	05(20)	07(28)	05(20)	07(28)	07(28)	07(28)	07(28)	05(20)	05(20)
100	Maintains	20(80)	20(80)	20(80)	22(88)	22(88)	20(80)	22(88)	20(80)	20(80)	20(80)	20(80)	22(88)	22(88)
	Conservative	05(20)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)
	Liberal	00(00)	05(20)	05(20)	03(12)	03(12)	05(20)	03(12)	05(20)	05(20)	05(20)	05(20)	03(12)	03(12)
200	Maintains	25(100)	22(88)	22(88)	25(100)	25(100)	22(88)	25(100)	22(88)	22(88)	22(88)	22(88)	25(100)	25(100)
	Conservative	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)	00(00)
	Liberal	00(00)	03(12)	03(12)	00(00)	00(00)	03(12)	00(00)	03(12)	03(12)	03(12)	03(12)	00(00)	00(00)

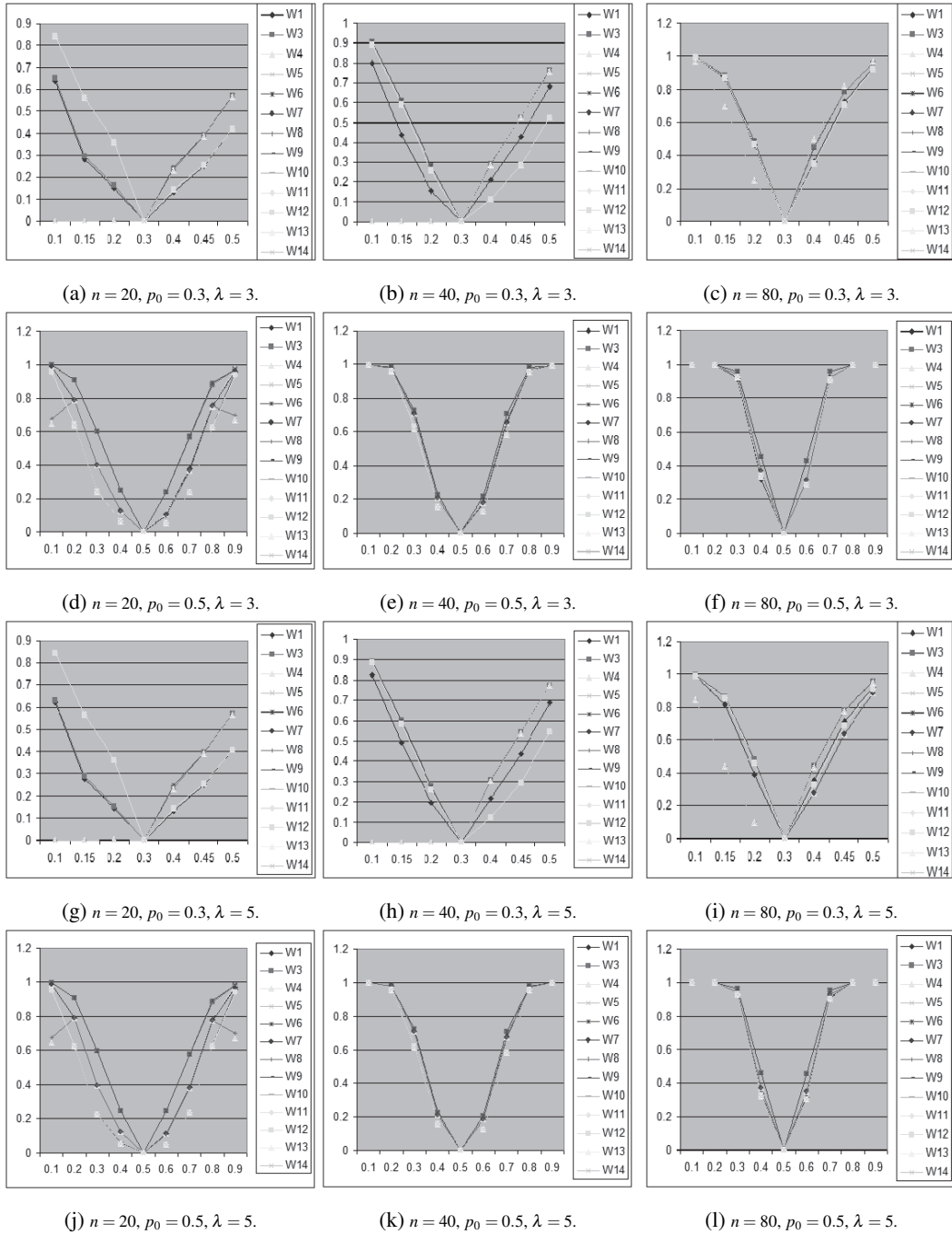


Figure 1: Power functions of the tests for different values of n , p_0 and λ .

Table 5: Powers of the tests for $n = 20$, $p_0 = 0.3$, $\lambda = 3$.

p	W_1	W_3	W_4	W_5	W_6	W_7	W_8	W_9	W_{10}	W_{11}	W_{12}	W_{13}	W_{14}
0.10	0.6380	0.6497	0.8453	0.8453	0.8453	0.8453	0.0006	0.8453	0.8453	0.8453	0.8453	0.0006	0.0006
0.15	0.2808	0.2973	0.5634	0.5635	0.5635	0.5634	0.0007	0.5634	0.5634	0.5634	0.5634	0.0007	0.0007
0.20	0.1492	0.1635	0.3584	0.3602	0.3602	0.3584	0.0026	0.3584	0.3584	0.3584	0.3584	0.0026	0.0026
0.30	0	0	0	0	0	0	0	0	0	0	0	0	0
0.40	0.1331	0.2354	0.1469	0.2462	0.2462	0.1464	0.2301	0.1469	0.1462	0.1462	0.1462	0.2298	0.2298
0.45	0.2542	0.3921	0.2576	0.3912	0.3912	0.2576	0.3874	0.2576	0.2576	0.2576	0.2576	0.3874	0.3874
0.50	0.4197	0.5717	0.4212	0.5689	0.5689	0.4212	0.5676	0.4212	0.4212	0.4212	0.4212	0.5675	0.5675

Table 6: Powers of the tests for $n = 40$, $p_0 = 0.3$, $\lambda = 3$.

p	W_1	W_3	W_4	W_5	W_6	W_7	W_8	W_9	W_{10}	W_{11}	W_{12}	W_{13}	W_{14}
0.10	0.7991	0.9052	0.8932	0.9052	0.9052	0.8932	0.0001	0.8932	0.8932	0.8932	0.8932	0.0001	0.0001
0.15	0.4365	0.6107	0.5896	0.6112	0.6112	0.5896	0.0001	0.5896	0.5896	0.5896	0.5896	0.0001	0.0001
0.20	0.1543	0.2805	0.2603	0.2806	0.2806	0.2603	0.0004	0.2603	0.2603	0.2603	0.2603	0.0004	0.0004
0.30	0	0	0	0	0	0	0	0	0	0	0	0	0
0.40	0.2110	0.2889	0.1120	0.2892	0.2892	0.1120	0.2873	0.1117	0.1117	0.1117	0.1117	0.2873	0.2873
0.45	0.4288	0.5273	0.2855	0.5277	0.5277	0.2855	0.5251	0.2854	0.2854	0.2854	0.2854	0.5249	0.5249
0.50	0.6785	0.7584	0.5245	0.7590	0.7590	0.5245	0.7574	0.5243	0.5243	0.5243	0.5243	0.7574	0.7574

Table 7: Powers of the tests for $n = 80$, $p_0 = 0.3$, $\lambda = 3$.

p	W_1	W_3	W_4	W_5	W_6	W_7	W_8	W_9	W_{10}	W_{11}	W_{12}	W_{13}	W_{14}
0.05	0.9937	0.9942	0.9939	0.9941	0.9941	0.9939	0.9693	0.9939	0.9939	0.9939	0.9939	0.9693	0.9693
0.10	0.9905	0.9940	0.9938	0.9941	0.9940	0.9938	0.8463	0.9939	0.9939	0.9939	0.9939	0.8463	0.8463
0.15	0.8652	0.8804	0.8697	0.8783	0.8783	0.8697	0.6951	0.8697	0.8697	0.8697	0.8697	0.6951	0.6951
0.20	0.4621	0.4896	0.4696	0.4863	0.4863	0.4696	0.2479	0.4696	0.4696	0.4696	0.4696	0.2480	0.2480
0.25	0.3872	0.4783	0.4622	0.4748	0.4748	0.4622	0.0964	0.4622	0.4622	0.4622	0.4622	0.0964	0.0964
0.30	0	0	0	0	0	0	0	0	0	0	0	0	0
0.35	0.3510	0.4415	0.2786	0.4415	0.4415	0.2786	0.4376	0.3310	0.3310	0.3310	0.3310	0.4376	0.4376
0.40	0.3628	0.4497	0.3495	0.4497	0.4497	0.3495	0.4504	0.3495	0.3495	0.3495	0.3495	0.4979	0.4979
0.45	0.7228	0.7888	0.7081	0.7888	0.7888	0.7081	0.7892	0.7081	0.7081	0.7081	0.7081	0.8197	0.8197
0.50	0.9221	0.9539	0.8863	0.9539	0.9539	0.8863	0.9524	0.9129	0.9129	0.9129	0.9129	0.9524	0.9524
0.55	0.9310	0.9550	0.9240	0.9550	0.9550	0.9240	0.9550	0.9240	0.9240	0.9240	0.9240	0.9654	0.9654

6. Discussion and Conclusion

In this paper, the comparisons are made between the performances of the proposed tests for the cured proportion in a recurrent event count data set when the data are censored. The comparisons were with respect to type I error rates and the power of the tests. The likelihood ratio test maintains type I error rate in more number of configurations when the sample sizes are 20 and 40 compared to the remaining tests. The tests which follow the LRT are the variants of the Wald test and that of the score test. But as the sample size increased, the performances of the variants of the Wald test, the score test and its variants were on par with the likelihood ratio test. The power comparisons of these tests are examined only when they maintain type I error rate. In these cases, the powers of the likelihood ratio test are lower compared to that of the Wald test, the score test and their variants. The newly proposed variants of the score test remain at par with the standard score test. The computation of these new variants is much simpler compared to the usual score test. As in the case of uncensored data set up, the score test and all its variants exhibit fluctuations in their power functions. The results of the simulations indicate that it is advisable to take a minimum sample of size 100. Based on the size and power of the tests, the variants of the Wald test viz., W_5 and W_6 can be recommended to the users when testing for the cured proportion in a censored data set up. Moreover, it was also observed that censoring has not much impact on the power of the tests but, the number of times the tests maintain type I error rate is less when compared to that of the uncensored case (Sumathi and Rao, 2010).

As mentioned by one of the referees, when testing for $p = 0$ or $p = 1$, the asymptotic distribution of the LRT is not a chi-square distribution with one degree of freedom. (See Self and Liang (1987), Maller and Zhou (1996)) We tried to simulate the null distribution of the test statistics at $p = 0$ and at $p = 1$. The Newton-Raphson procedure did not converge in many instances and hence we did not perceive the idea. This difficulty was also encountered when testing for small values of the inflate parameter such as $p = 0.1$ and large values of parameter such as $p = 0.7$ and $p = 0.9$. This issue could not be resolved even when the simulations were increased considerably.

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A The Marginal Likelihood Function when the Censoring Time follows Weibull Distribution.

The probability distribution function of a random variable t_i following Weibull distribution is $f(t_i; \theta, \alpha) = (\theta\alpha)(\theta t_i)^{\alpha-1} (e^{-\theta t_i})^\alpha$, $t_i > 0, \theta, \alpha > 0$.

The contribution from the i^{th} observation to the conditional likelihood is

$$L_i = \left(p + (1-p)e^{-\lambda t_i} \right)^{\delta_i} \left(\frac{(1-p)e^{-\lambda t_i} (\lambda t_i)^{y_i}}{y_i!} \right)^{(1-\delta_i)} \quad (23)$$

where $\delta_i = \begin{cases} 1, y_i = 0 \\ 0, y_i \neq 0 \end{cases}$ and t_i 's, $i = 1, 2, \dots, n$, are i.i.d. Weibull random variables with density

function $f(t_i; \theta, \alpha)$.

Case 1: When $T = \infty$, the contribution of the i^{th} observation to the marginal likelihood is

$$M_1 = \int_0^\infty (p + (1-p)e^{-\lambda t_i}) f(t_i; \theta, \alpha) dt_i, y_i = 0 \quad \text{and}$$

$$M_2 = \int_0^\infty \frac{(1-p)e^{-\lambda t_i} (\lambda t_i)^{y_i}}{y_i!} f(t_i; \theta, \alpha) dt_i, y_i > 0.$$

On integration, we get

$$M_1 = \left(p \left(\frac{1}{\alpha} \right)^{\alpha-1} + (1-p) \left(\frac{\theta \alpha}{\lambda + \theta \alpha} \right) \left(\frac{\theta}{\lambda + \theta \alpha} \right)^{\alpha-1} \right) \Gamma \alpha, y_i = 0 \quad \text{and}$$

$$M_2 = (1-p) \alpha \left(\frac{\theta}{\lambda + \theta \alpha} \right)^\alpha \left(\frac{\lambda}{\lambda + \theta \alpha} \right)^{y_i} \frac{\Gamma(y_i + \alpha)}{\Gamma(y_i + 1)}, \quad y_i > 0.$$

The full marginal likelihood is $L(p, \lambda, \theta, \alpha; y) = \prod_{i=1}^n M_1^{\delta_i} M_2^{1-\delta_i}$ which simplifies to

$$\begin{aligned} L(p, \lambda, \theta, \alpha; \underline{y}) &= \prod_{i=1}^n \left[\left\{ p \alpha^{\alpha-1} + (1-p) \left(\frac{\theta}{\lambda + \theta \alpha} \right)^\alpha \alpha \Gamma \alpha \right\}^{\delta_i} \right. \\ &\quad \left. \left\{ (1-p) \alpha \left(\frac{\theta}{\lambda + \theta \alpha} \right)^\alpha \left(\frac{\lambda}{\lambda + \theta \alpha} \right)^{y_i} \frac{\Gamma(y_i + \alpha)}{\Gamma(y_i + 1)} \right\}^{1-\delta_i} \right] \end{aligned} \quad (24)$$

Case 2: When T is finite, i.e., say $T = c$, the contribution of the i^{th} observation to the marginal likelihood is

$$\begin{aligned} M_3 &= \int_0^c (p + (1-p)e^{-\lambda t_i}) \frac{f(t_i; \theta, \alpha)}{\int_0^c f(t_i; \theta, \alpha) dt_i} dt_i, y_i = 0 \\ &= \int_0^c (p + (1-p)e^{-\lambda t_i}) \frac{(\theta \alpha) (\theta t)^{\alpha-1} (e^{-\alpha \theta})^\alpha}{(\theta c)^{\alpha-1} e^{-\alpha \theta c} + \alpha (\alpha-1) \theta^\alpha \int_0^c t_i^{\alpha-2} e^{-\alpha \theta t_i} dt_i} dt_i \end{aligned}$$

and

$$\begin{aligned} M_4 &= \int_0^c \frac{(1-p)e^{-\lambda t_i} (\lambda t_i)^{y_i}}{y_i!} \frac{f(t_i; \theta, \alpha)}{\int_0^c f(t_i; \theta, \alpha) dt_i} dt_i, y_i > 0 \\ &= \int_0^c \frac{(1-p)e^{-\lambda t_i} (\lambda t_i)^{y_i}}{y_i!} \frac{(\theta \alpha) (\theta t)^{\alpha-1} (e^{-\alpha \theta})^\alpha}{(\theta c)^{\alpha-1} e^{-\alpha \theta c} + \alpha (\alpha-1) \theta^\alpha \int_0^c t_i^{\alpha-2} e^{-\alpha \theta t_i} dt_i} dt_i \end{aligned}$$

The full marginal likelihood $L(p, \lambda, \theta, \alpha; \underline{y}) = \prod_{i=1}^n M_3^{\delta_i} M_4^{1-\delta_i}$ does not have a closed form.